Analysis & Design of Algorithms.

Assignment-1

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Sem-4

Ans-1 Time complexity = $O(n)$ because each recursion follow $T(n-1)$.
Space complexity = $O(n)$ because recursive call stack of depth n.

Ans-2b Best case $O(1)$ → key is found on 1st comparison.
worst case $O(n)$ → If key is not present or at last pos.
Average case $O(n)$ → If key is at any mid pos.
Space complexity $O(1)$ → Because only one variable i is used.

Ans 1c Recurrence Relation (func. 1).

(1) $T(n) = T(n-1) + 1$ → (1)

For $(n-1)$

$T(n-1) = T(n-2) + 1$ → (2)

Put (2) in (3)

$T(n) = T(n-2) + 1 + 1$

$T(n) = T(n-2) + 2$

$\vdots$

$T(1) + (n-1)$

$T(n) = O(n)$.

Space complexity = $O(n)$

② func. 2.

Recurrence Relation

$T(n) = 2T(n-1)$. → ①

for $T(n-1) = 2T(n-2)$. → ②

Put ② in ①

$T(n) = 2(2T(n-2))$

$T(n) = 4T(n-2)$

$= 2^2 T(n-2)$.

$\vdots$

$2^3 T(n-3)$

$\vdots$

~~$2^n T(n-2)$~~.

$= 2^n T(0)$

$T(n) = O(2^n)$.

space complexity $= O(n)$

Ans-2 ① Best case

$T(n) = 3n+5$.

Order = n no. of times.

$\therefore O(n)$.

② Average case

$T(n) = n^2+4$

Order = n^2.

$\therefore O(n^2)$.

③ worst case.

$T(n) = 2n^3+n$.

Order = n^3

$\therefore O(n^3)$.

Ans-3. $[\underbrace{5}_{S.P}, \underbrace{2, 4, 6, 1, 3}_{U.P.}]$.

Sorted Part = 5.

comp. 2 with 5.

$2 < 5$.

∴ swap.

$[\underbrace{2, 5}_{S.P}, \underbrace{4, 6, 13}_{U.P.}]$.

comp. 4 with 5 & 2.

$4 < 5$, $4 > 2$

∴ swap.

$[\underbrace{2, 4, 5}_{S.P}, \underbrace{6, 1, 3}_{U.P.}]$

comp. 6 with. 2, 4, 5.

∴ NO swap

[2, 4, 5, 6, 1, 3

comp. 1 with. 2, 4, 5, 6.

$1 < 6$, $1 < 5$, $1 < 4$, $1 < 2$.

∴ swap

$[\underbrace{1, 2, 4, 5, 6}_{S.P}, \underbrace{3}_{U.P.}]$

∴ last swap.

[1, 2, 3, 4, 5, 6]

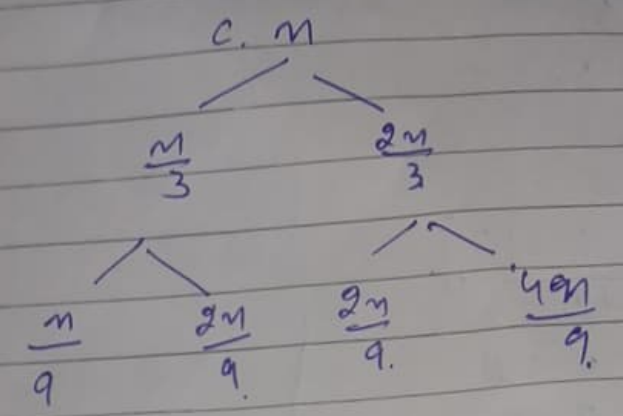
∴ $T(n) = O(n)$. n no. of ~~times~~ elements (best case)

worst case = $O(n^2)$

Avg. case = $O(n^2)$

space complexity = $O(1)$.

Ans-4. $T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right)$. $\frac{1}{3}$ & $\frac{2}{3}$.



$\frac{1}{3} \cdot \frac{2}{3}$

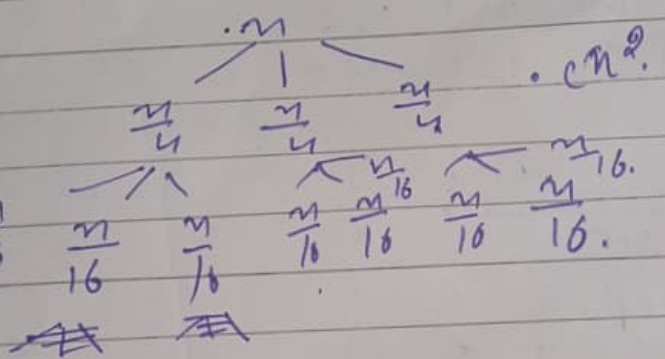
$\frac{2}{3} \cdot \frac{1}{3}$

$\frac{2n}{3} \cdot \frac{2}{3}$

$c\left(\frac{n}{9}\right) + c\left(\frac{2n}{9}\right) + c\left(\frac{2n}{9}\right) + c\left(\frac{4n}{9}\right) = cn.$

$\therefore O(n\log n)$.

Ans-5 $T(n) = 3T\left(\frac{n}{4}\right) + O(n^2)$.



$\cdot cn^2$.

$\left(\frac{n}{4}\right)^2 = \frac{n^2}{16} = \left(\frac{n}{16}\right)^2$.

$cn^2 + \frac{3}{16}cn^2 + \left(\frac{3}{16}\right)^2 cn^2 + \left(\frac{3}{16}\right)^3 \cdots$

$cn^2\left[1 + \frac{3}{16} + \left(\frac{3}{16}\right)^2 \cdots\right]$.

$cn^2\left[\frac{1}{1-\frac{3}{16}}\right] = cn^2 = O(n^2)$

Ans 6 ① $T(n) = 2T\left(\frac{n}{2}\right) + n$.

$a = 2 \quad b = 2 \quad p = 0 \quad k = 1$.

$a = b^k$.

$2 = 2^1$

$\therefore T(n) = \Theta(n^{\log_2 2} \log^{p+1} n)$. As $\log 2 = 1$.

$= \Theta(n \log n)$.

② $T(n) = 4T\left(\frac{n}{2}\right) + n^2$.

$a = 4 \quad b = 2 \quad k = 2. \quad p = 0$.

$a = b^k$

$4 = 2^2$

$\therefore T(n) = \Theta(n^{\log_b a} \log^{p+1} n)$.

$= \cancel{\Theta(n \log}$

$= \Theta(n^{\log_2 4} \log^{0+1} n)$.

$= \Theta(n^2 \log n)$.

③. $T(n) = 3T\left(\frac{n}{2}\right) + n$.

$a = 3 \quad b = 2. \quad k = 1 \quad p = 0$.

$a = b^k$

$3 \Rightarrow 2$.

$\therefore T(n) = \Theta(n^{\log_b a})$.

$= \Theta(n^{\log_2 3})$.